Collections recovery: the “+11% month-on-month” claim is not real

Executive memo · data: 1 Jan – 8 Aug 2026 (the file holds ≈7.3 months, not 12) · every figure is reproducible from the repository

1. What happened

The 11% is reproducible — and it is an artefact. Summing raw SUCCESS payment rows gives Feb→Mar **+11.0%**. On the cleaned (golden) data total recovery rose +10.5%, but March has 31 days and February 28 (**+10.7%** from the calendar alone). **Per day, recovery moved -0.22%** (95% CI -6.7% to +6.7%). Daily recovery is flat across Jan–Jul (≈₹55 lakh/day, no break-point, permutation p=0.76). Contact, right-party-contact, PTP-kept and recovery-rate show no significant Jan–Jul trend (all trend p>0.3); the largest single-month move is 2.2 pp. The one drift is *downward*: PTP rate 30.0% → 27.8% (nominal p=0.013, FDR q=0.11) — a watch item, not a finding.

Money that is not recovery: of ₹191.7 Cr in raw payment rows only **₹119.8 Cr** is settled, de-duplicated recovery on delinquent accounts (duplicates ₹2.4 Cr = 1.9% of Jan–Jul recovery; non-settled FAILED/PENDING/REVERSED ₹56.3 Cr; payments on current accounts ₹11.8 Cr).

2. Why — and what is not the cause

Calendar length and **definitions** explain the headline. We tested portfolio mix, DPD, geography, risk segment, agent, tenure, campaign, channel, vendor, calling hour/weekday and attempt frequency: **2 of 29 tests look significant before multiple-testing correction and 0 after it**. No Simpson reversals in 42 scans; mix effect on Feb→Mar is ≈0. Client and language are not in the data. Payment-to-touch “attribution” is chance: 11.9% of payments follow a contact within 7 days vs 12.3% for random dates. Adding more touches does not raise next-period payment (within-account slopes ≈0, all FDR q>0.9).

Integrity red flags: 23% of recovery ₹ lands on accounts that status history shows as PAID/CLOSED/WRITEOFF; PTP “KEPT” is barely corroborated by payments (5.0% vs 4.1% for BROKEN); a rate computed with all payments over the shrinking “active” base would show a fake +56% gain; 54% of calls are time-stamped outside 08:00–19:00 IST (compliance exposure if real); agent IDs cannot be reconciled.

3. How confident are we?

Conclusion	Class
The 11% is calendar + definitions; per-day recovery is flat	Fact (identity) / Strong evidence (CI, no break-point)
Duplicate payments overstate recovery by 1.9%	Fact
No tested driver explains differences; payment-touch link = chance	Strong evidence of no large effect (small effects undetectable)
More attempts / WhatsApp / field visits do not lift payment	Correlation (observational; reverse causality possible)
Effect of the mid-year targeting change (DiD +10% [-10,+30] at Jul-1; -4% at Apr-1)	Hypothesis — not identifiable
Compliance and agent-level conclusions	Hypothesis (timestamp / identity quality)

4. What should we do with ₹10 Cr?

Recommendation: Option 4 — better borrower targeting — released in tranches, each gated by a randomised holdout. It is not chosen because its payoff is proven; it is chosen because it is the only option whose effect can be measured cheaply and the data show no marginal return to capacity: telephony vendors/hours do not differ in contact rate, extra calls/agents/WhatsApp/field visits show no response, there is no AI-voice data and no cost data for any option. The data foundation the investment must include (as-of status, valid account↔borrower↔call keys, one agent identity) is also the prerequisite for evaluating *any* future spend.

Expected financial impact. Baseline net recovery is ₹199 Cr/year (95% CI ₹195–203 Cr). **Break-even = +5.0% recovery** in year 1 if each recovered rupee counts as benefit (+10% at 50% capture, +20% at 25%). **Expected incremental recovery is not reliably estimable** from this data; the evidenced central case is ₹0. Scenarios (100% capture, 12 months, all-in cost ₹10 Cr): 0% uplift → ROI −100%; +5% → ₹10 Cr, ROI 0%; +10% → ₹20 Cr, ROI +99%; +15% → ROI +199%. Implementation/operating cost cannot be built from the data (none supplied): ₹10 Cr is treated as all-in; opex beyond it would raise break-even. **Downside:** if uplift is 0 the full ₹10 Cr is lost — hence tranches.

Pilot design (to turn the hypothesis into an estimate). Randomise 5,463 of 27,315 delinquent accounts (20%) to a never-treated holdout, stratified by DPD bucket × outstanding band; outcome = ₹ recovered per account, measured per observed day; pre-register the metric. A 20% holdout over 3 months detects a ≥4.8% relative effect (80% power), i.e. just below the 5.0% break-even; ~4,976 accounts per arm are needed for 5%. Release tranche 2 only if the lower CI bound clears break-even; stop if the point estimate is ≤0.

5. Immediate actions

(1) Retire the 11% figure; report **net recovery per observed day** and rates in **percentage points**. (2) Adopt the golden definitions and DQ gates (de-duplicate on payment_id, settled-only, delinquent-only, one clock). (3) Fix status history and the call↔attempt↔disposition keys. (4) Obtain cost per channel/agent-hour and the strategy-change date/assignment rule to enable cost-per-₹ and a proper counterfactual.

Limitations: single account snapshot (no as-of DPD); timestamps without a timezone assumed IST (robustness checked); several fields behave as independent random draws, so absence of effects is reported as absence of *detectable* effects.